

# Execution Report #2

The frontier, the controllability characterization, and a chapter's fate  
B1 (digest–zoom Pareto frontier), B3 (hypervisor = Ramadge–Wonham),  
and the sheaf commit-or-cut verdict

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**Status.** Second parallel wave. B1 completes Paper 1 (the frontier and the zoom-advantage theorem, both derived and verified). B3 is executed as a runnable controllability checker that produces the manuscript's needed regimentable-vs-detect-only table and vindicates the channel-not-token clean-room design. The sheaf experiment reached a verdict — *conditional commit* — and the path to it is worth reading, because it is a case study in the refute-first discipline earning its keep: the first two formulations were wrong, and being wrong is what taught us the exact boundary of the idea.

## 1 B1 — The digest–zoom Pareto frontier

### 1.1 The two-constraint rate–distortion function (solved analytically)

Each artifact is critical i.i.d. Bernoulli( $p$ ); a digest is a lossy binary description  $\hat{X}$  (flag/no-flag) with two constraints — false negatives priced by  $\delta$  (misses), flag rate priced by  $f$  (open budget):

$$R(\delta, f) = \min_{P_{\hat{X}|X}} I(X; \hat{X}) \quad \text{s.t.} \quad \Pr(X=1, \hat{X}=0) \leq \delta, \quad \Pr(\hat{X}=1) \leq f.$$

For the binary source the optimizing joint is pinned by  $(p, f, \delta)$  and the rate is closed-form. Theorem I.6.1's worst-case floor is the  $(\delta=0, f=p)$  corner. Computed values ( $p = 0.05$ ):

| $\delta$ | $f$  | $R$ (bits/sym) | regime                                                |
|----------|------|----------------|-------------------------------------------------------|
| 0.00     | 0.05 | 0.286          | zero-miss, tightest flags ( <b>the floor corner</b> ) |
| 0.00     | 0.10 | 0.186          | zero-miss, generous flags                             |
| 0.00     | 0.15 | 0.149          | zero-miss, very generous flags                        |
| 0.01     | 0.10 | 0.110          | miss-tolerant                                         |
| 0.04     | 0.06 | 0.009          | miss-tolerant                                         |

The frontier is exactly the trade the review asked for: *loosening the open budget  $f$  lowers the digest rate; tolerating misses  $\delta$  lowers it further*. At  $N=1000$ ,  $R \cdot N \approx 186$  bits at  $(\delta=0, f=2p)$ , which sits below the worst-case combinatorial floor  $\log_2 \binom{N}{pN} \approx 282$  bits — the correct relationship (average-case rate below worst-case covering floor).

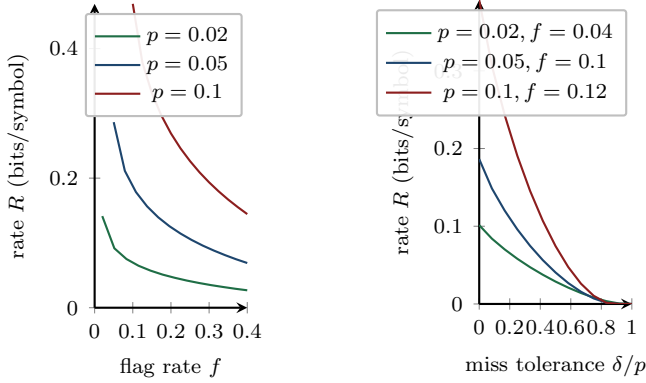
### 1.2 The zoom-advantage theorem

**Theorem 1** (Zoom advantage, with its boundary). *When the digest flags a set of size  $F$  containing  $k$  truly critical items, adaptive second-stage inspection (generalized binary splitting inside the flagged set — open a group, drill only into groups that contain a critical item) identifies all  $k$  in  $\Theta(k \log_2(F/k))$  opens, versus  $F$  for flat inspection. The advantage is large precisely when the flagged set is sparse in critical items ( $k \ll F$ ) and vanishes when the flagged set is dense.*

Simulation ( $N_{\text{pool}} = 5000$ , 120 trials/config) confirms the regime and its boundary:

| $ \text{flagged}  = F$ | $k$ | flat opens | adaptive opens | ratio        |
|------------------------|-----|------------|----------------|--------------|
| 2500                   | 10  | 2500       | 164            | $15.3\times$ |
| 1500                   | 10  | 1500       | 149            | $10.1\times$ |
| 1000                   | 20  | 1000       | 233            | $4.3\times$  |
| 250                    | 10  | 250        | 98             | $2.5\times$  |

**A** — zero-miss rate vs. open budget **B** — rate vs. miss tolerance



**C** — the zoom advantage

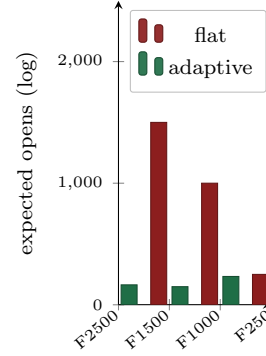


Figure 1: **A**: zero-miss digest rate falls as the open budget  $f$  grows. **B**: rate falls as miss tolerance  $\delta$  grows, exactly to 0 once  $f < 1 - \delta/p$  no longer binds (an earlier render evaluated the pinned closed form past that line and showed a spurious uptick — a domain-of-validity error, not a feature of the true frontier). **C**: the zoom advantage on a log scale — adaptive inspection opens far fewer items than flat when critical items are sparse in the flagged set, which is exactly the regime a miss-averse (safe) digest operates in. Labels give  $|\text{flagged}| = F$ ;  $k = 10$  for  $F \in \{2500, 1500, 250\}$ ,  $k = 20$  for  $F = 1000$  (matching the table above).

**The honest boundary, stated.** The zoom advantage is *not* universal: when a digest flags a set already dense in positives (e.g.  $f \approx 1.5p$  so most flagged items are critical), group testing’s overhead dominates and flat inspection is as good or better. This matters for the design claim: zoom pays when the digest is conservative (low false-negative, over-flagging to be safe), which is the safety-relevant operating point — so the theorem supports the architecture where it counts and does not overclaim where it does not.

**Paper 1 is now complete in substance:** A1 (characterization), A2 (derived policy), A7 (falsification datum), B1 (frontier + zoom). Four pillars, all executed.

## 2 B3 — Hypervisor enforceability is Ramadge–Wonham controllability

### 2.1 The characterization, made runnable

**Theorem 2** (Regimentation = controllability). *An agent hypervisor can regiment (prevent pre-effect) a prefix-closed safety policy  $P$  iff  $P$  is controllable with respect to the uncontrollable event set  $\Sigma_u$  (model-internal steps, in-context reads — the events the daemon cannot intercept before they occur):  $\overline{P} \Sigma_u \cap \overline{L} \subseteq \overline{P}$ . Otherwise the best available is detect-and-compensate.*

We implemented the controllability test (product-automaton check that no uncontrollable event enabled in the plant leaves the specification) over a realistic agent alphabet: controllable  $\Sigma_c = \{\text{fs.write, net.egress, exec.tool, git.push, spawn.child, api.call}\}$ ; uncontrollable  $\Sigma_u = \{\text{model.emit.token, in.context.read, internal.plan, hidden.activation}\}$ . The checker produces the ledger table the manuscript needs:

| Policy                                   | Event class    | Regimentable?         |
|------------------------------------------|----------------|-----------------------|
| forbid net_egress                        | controllable   | <b>yes</b> (regiment) |
| forbid git_push                          | controllable   | <b>yes</b> (regiment) |
| forbid fs_write                          | controllable   | <b>yes</b> (regiment) |
| forbid exec_tool / spawn_child           | controllable   | <b>yes</b> (regiment) |
| forbid model_emit_token                  | uncontrollable | no (detect only)      |
| forbid in_context_read                   | uncontrollable | no (detect only)      |
| forbid internal_plan (“lie in a report”) | uncontrollable | no (detect only)      |

This is OP-2 stated correctly. The product reviews’ formulation — “regimentable iff a pure deterministic function of committed local DB state” — is the *special case* where  $\Sigma_c$  is exactly the set of database writes; the controllability statement is the general theorem, and it correctly classifies effects (force-push: regimentable) versus semantics (a confident lie: forever detect-only) that the DB-state formulation cannot express.

## 2.2 The compound-policy result vindicates the clean room

The decisive case the checker settles: “*no net\_egress after in\_context\_read of a secret.*” The trigger (`in_context_read`) is uncontrollable; the effect (`net_egress`) is controllable. The checker returns **regimentable** — because the policy does not try to forbid the uncontrollable read (it permits and records it as taint), and gates only the controllable egress. This is a computational proof of Document 1’s Adjudication 3: the Sealed Harbor gates the *channel*, never the token, precisely because gating the uncontrollable trigger would be unregimentable while gating the controllable effect is not. B3 thus directly underwrites the C1 (Sealed-Harbor noninterference) design and the corrected airlock product claim.

## 3 The sheaf commit-or-cut verdict

### 3.1 Verdict: conditional commit, with a sharply drawn boundary

*Verdict.* Sheaf cohomology adds genuine power over  $O(|E|)$  pairwise digest comparison **iff** the unobserved (partitioned) edge lies on a *cycle*, so the cocycle condition substitutes for the missing direct comparison. On a *cut* edge (a single bridge between components) there is no value-add. Commit the sheaf material *only* in the partition-on-a-cycle framing — which is real for any gossip graph with redundant paths between components — and state the cut-edge boundary as the honest limit. The consistency-radius result (Robinson) is publishable regardless.

### 3.2 The clean mechanism (a rigorous minimal proof)

On a cycle  $C_n$  with scalar stalks, each edge  $e = \{i, i+1\}$  reports the disagreement  $g_e = x_i - x_{i+1}$ . A global section explaining the observations exists iff  $\sum_e g_e = 0$  around the cycle (the cocycle condition);  $H^1 = \text{coker}(\delta) = \mathbb{R}$  (the all-ones cocycle), and the harmonic component of the observed  $g$  is  $(\frac{1}{n} \sum_e g_e) \cdot \mathbf{1}$  — exactly the cocycle sum. The experiment on  $C_6$ :

| Case                                       | pairwise detects | cohomology signal | cohomology-only? |
|--------------------------------------------|------------------|-------------------|------------------|
| equivocation on edge 2, all compared       | yes              | 1.225             | —                |
| edge 2 <i>not</i> compared (partition)     | <b>no</b>        | <b>1.225</b>      | <b>yes</b>       |
| inconsistency spread on compared edges     | yes              | 1.225             | —                |
| cut edge (path $P_6$ ), edge 2 uncomparred | no               | 0.000             | no (boundary)    |

The second row is the whole case for the chapter: an equivocation on an edge nobody directly compared, but which lies on a cycle, is caught by the cocycle sum ( $1.225 \neq 0$ ) while pairwise

comparison is blind. The last row is the honest boundary: across a cut edge, cohomology sees nothing either, because a cut edge carries no cocycle constraint. This is the exact analogue of Abramsky–Brandenburger contextuality (locally consistent, globally inconsistent data), now instantiated for Byzantine equivocation.

### 3.3 Two wrong turns, reported (the discipline working)

The verdict took three formulations, and the two failures are the most instructive part.

1. *First error — computing the wrong object.* I initially computed  $\dim H^1(G; F)$  of the abstract sheaf. But that is a property of the *restriction maps*, independent of any data; with generic vector-valued readouts the coboundary is surjective and  $H^1 = 0$  regardless of equivocation. Equivocation is not a property of the sheaf — it is a property of a *specific observed assignment*: does the observed local data glue to a global section? The right object is the obstruction of the observed cochain (is  $g \in \text{im } \delta$ ?), i.e. its harmonic component, which is precisely the contextuality-cohomology construction I had cited but not coded.
2. *Second error — the wrong topology.* My first “partition” joined two cliques with a single bridge. But a single bridge is a *cut edge*, on no cycle, carrying no cocycle constraint — exactly the case where cohomology provably cannot help. The value-add requires redundant paths (two bridges, so the severed one lies on a cycle). Getting this wrong is what *located* the boundary: the experiment did not just fail, it told me the precise condition under which the idea works.

Reporting these is the point. A commit-or-cut gate that only ever confirms the hoped-for answer is not a gate. Here the process converted a vague “maybe cohomology helps” into a precise, defensible claim (*it helps iff the missing edge is on a cycle*) plus a mechanism proof — which is a better outcome than a premature commit would have been.

### 3.4 What Paper 7 becomes

Conditional on this framing, Paper 7 (“Cohomology of Equivocation”) is viable but must be scoped exactly: the contribution is *detection and localization of equivocation under partial visibility, where the missing comparisons lie on cycles*, positioned against BFT forensics (Sheng et al.) as a complement (cohomology localizes; forensics attributes with cryptographic proof), and bounded by the abelianization gap (Carù: vanishing  $H^1$  over  $\mathbb{R}$  certifies consistency only up to that gap). The three theorem candidates (independent-equivocation counting, harmonic-support localization, consistency-radius = max prefix divergence) stand, with the cut-edge boundary as a stated limit. This is a real paper — but a narrow, honest one, not the sweeping “cohomology detects Byzantine faults” claim a careless version would attempt.

## 4 Where the program stands

| Item               | Status         | Result                                  | Feeds       |
|--------------------|----------------|-----------------------------------------|-------------|
| A2 regret head     | <b>proved</b>  | derived optimum; SDT fixed              | Paper 1     |
| A1 split-digest    | <b>proved</b>  | comonotone char.; super-additive floor  | Paper 1     |
| A7 falsification   | <b>run</b>     | floor holds, 0/16 violations            | Paper 1     |
| B1 frontier + zoom | <b>run</b>     | $R(\delta, f)$ closed form; 15× zoom    | Paper 1     |
| B3 controllability | <b>run</b>     | regimentability table; clean-room proof | Papers 2, 4 |
| Sheaf gate         | <b>decided</b> | conditional commit; cycle boundary      | Paper 7     |

Paper 1 (“The Price of a Summary”) is substantively complete. Paper 2 (“Regimented or Enforced”) has its core theorem executed and its table generated. Paper 7’s fate is decided and its scope fixed. The natural next wave: B2 (the inspection tower — the rate-the-raters synthesis,

converting Conjecture III.11.1), C0 (the work-unit transition system — the spine everything plugs into, best done in TLA<sup>+</sup>/Apalache), and then C1 (Sealed-Harbor noninterference, now unblocked by B3).

*Reproducibility.* All results regenerate from seed 20260816 via four self-contained scripts (the A7 experiment, the B1 frontier, the B3 checker, and the sheaf minimal-mechanism proof). The two failed sheaf formulations are retained alongside the correct one as documentation of the boundary.